Supplementary Material

LC-MS based urine untargeted metabolomic analyses to identify and subdivide urothelial cancer

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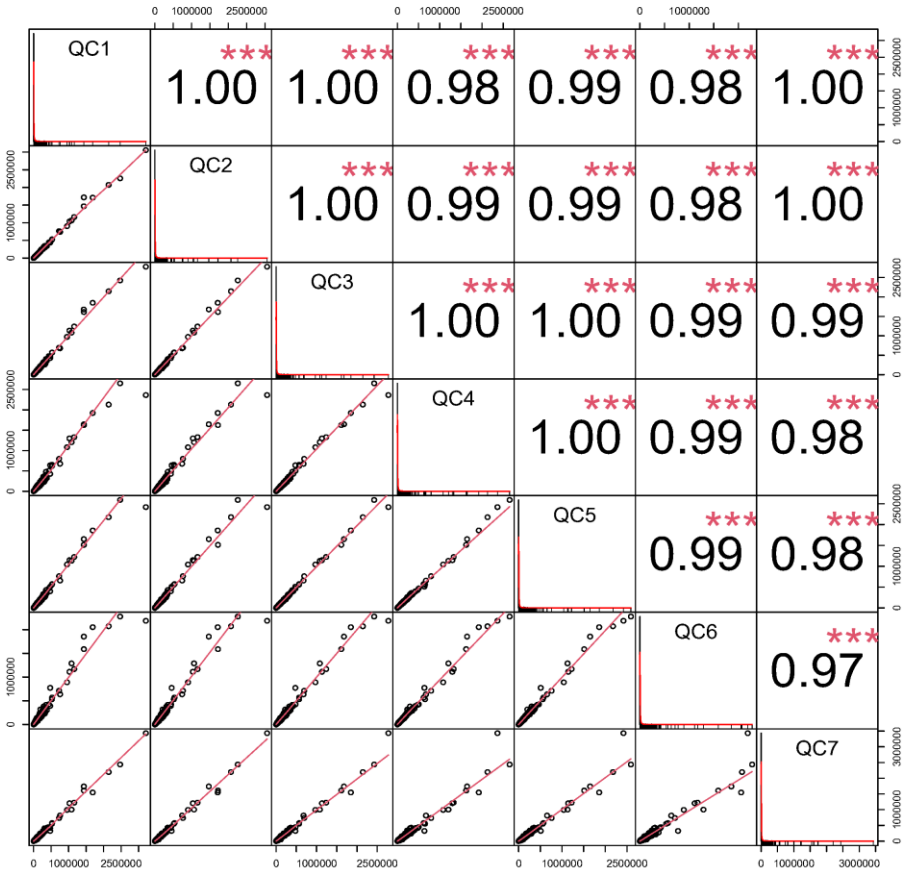
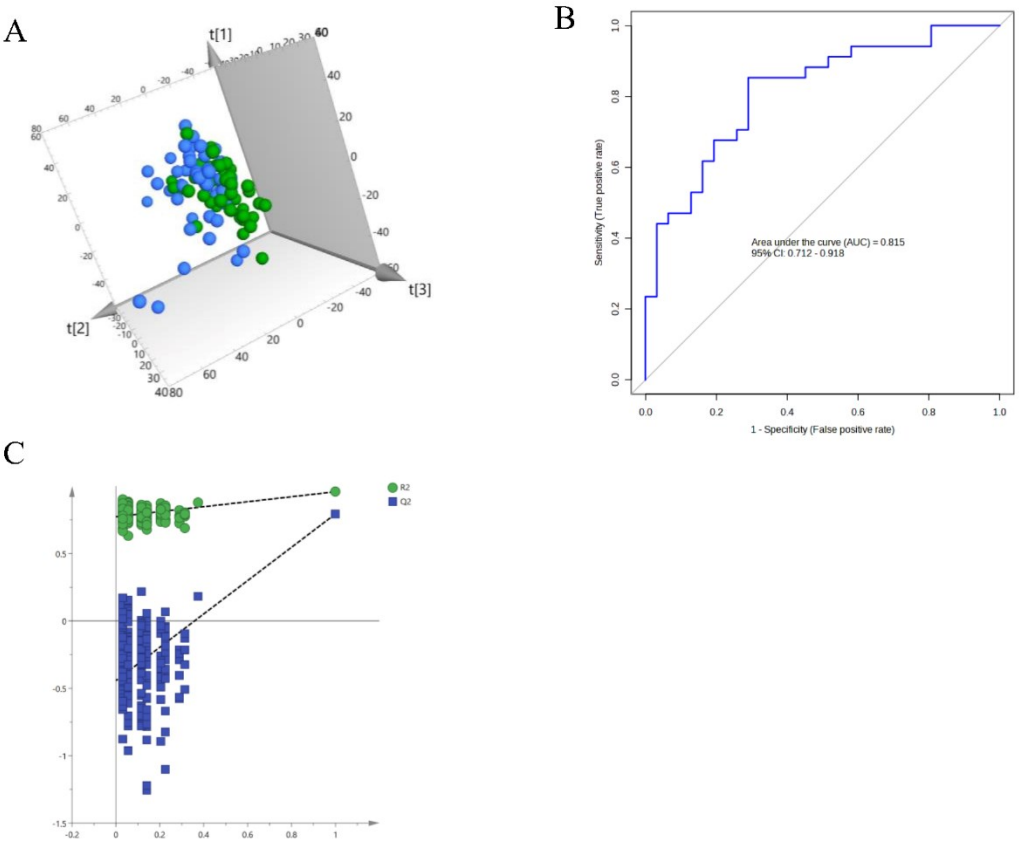
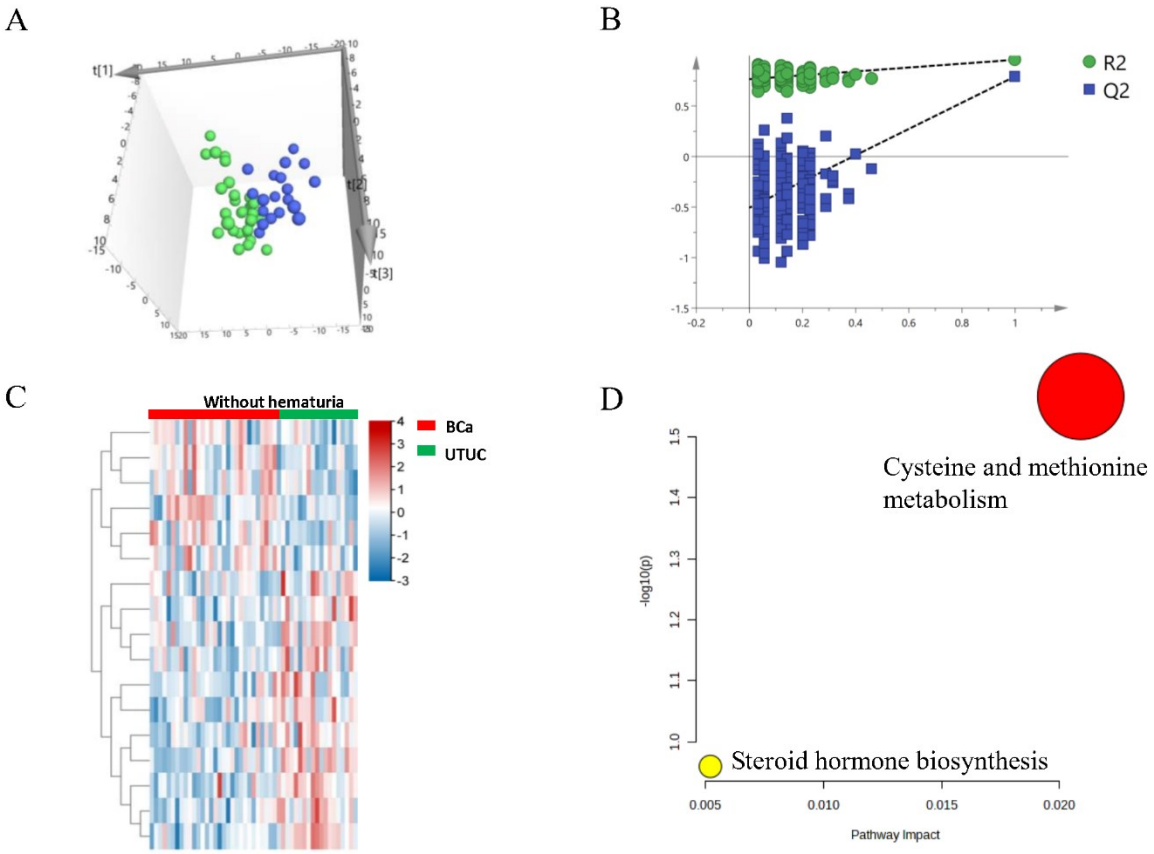
Supplementary Figure 1. | Assessment of QC samples.



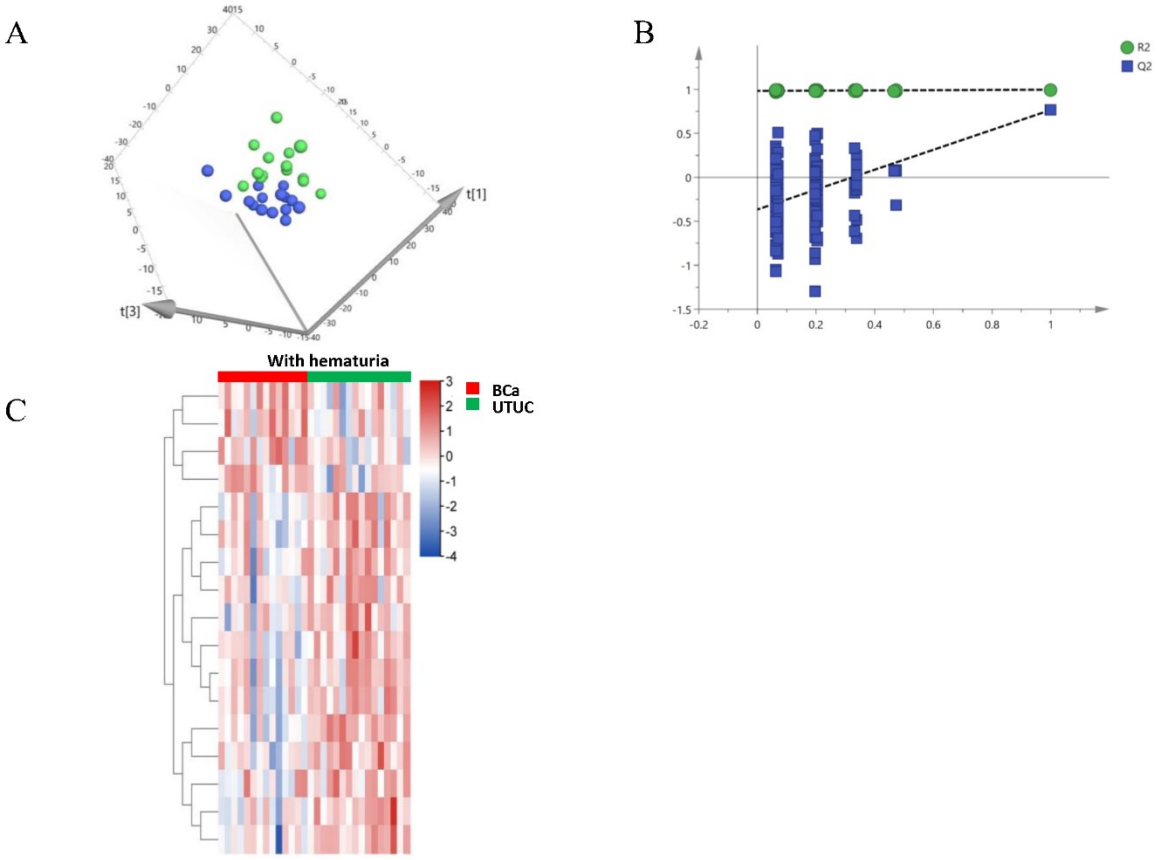
Supplementary Figure 2. | Analysis of metabolic profiling variation of UCs and HCs. **(A).** Score plot of supervised PCA overview of urine metabolic profiling between UCs and HCs. **(B).** ROC curve of the 10-fold cross validation in the discovery group based on the biomarker panel. **(C)**. Two hundred permutation tests of the OPLS-DA model based on UCs and HCs urine metabolome.



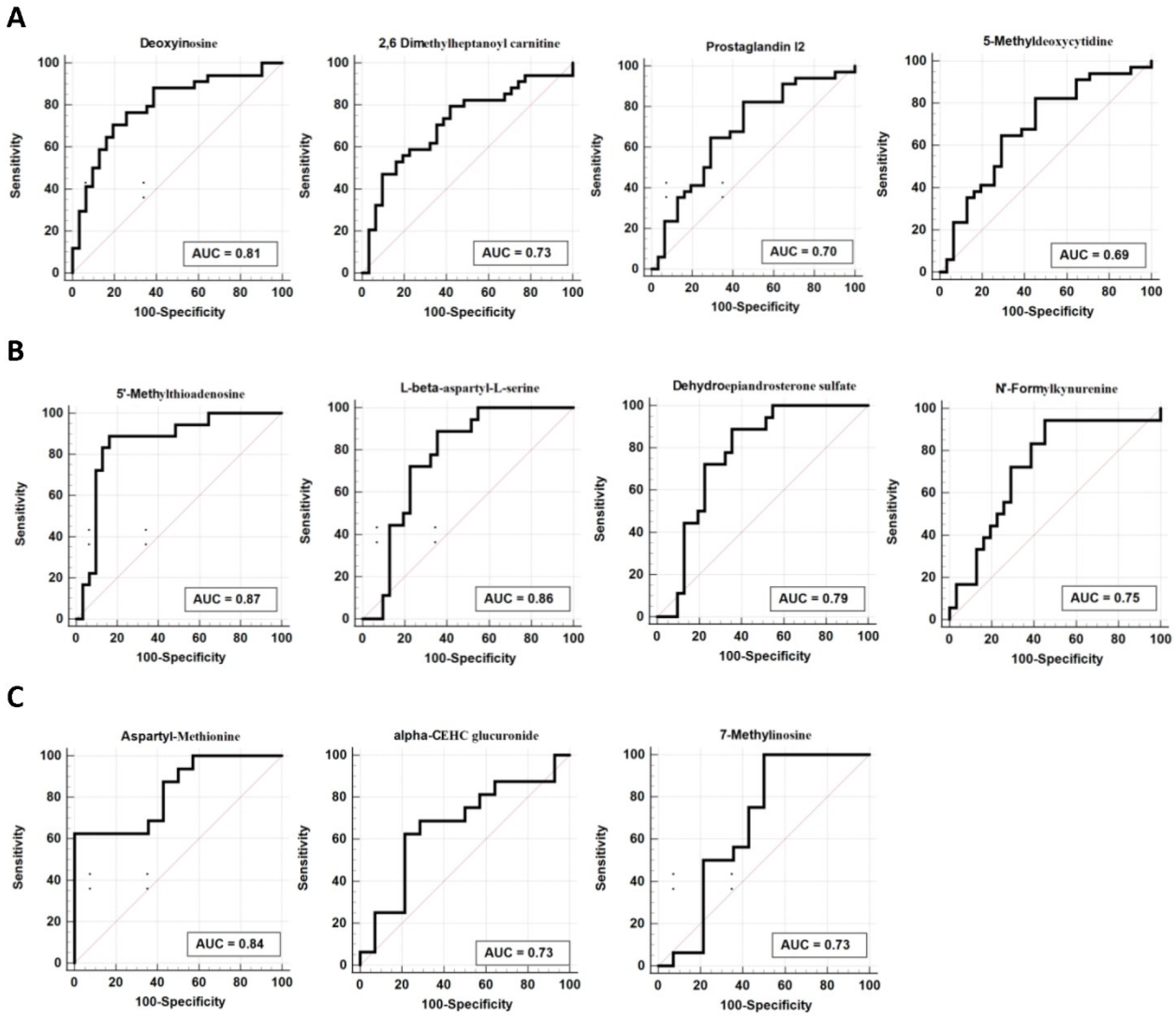
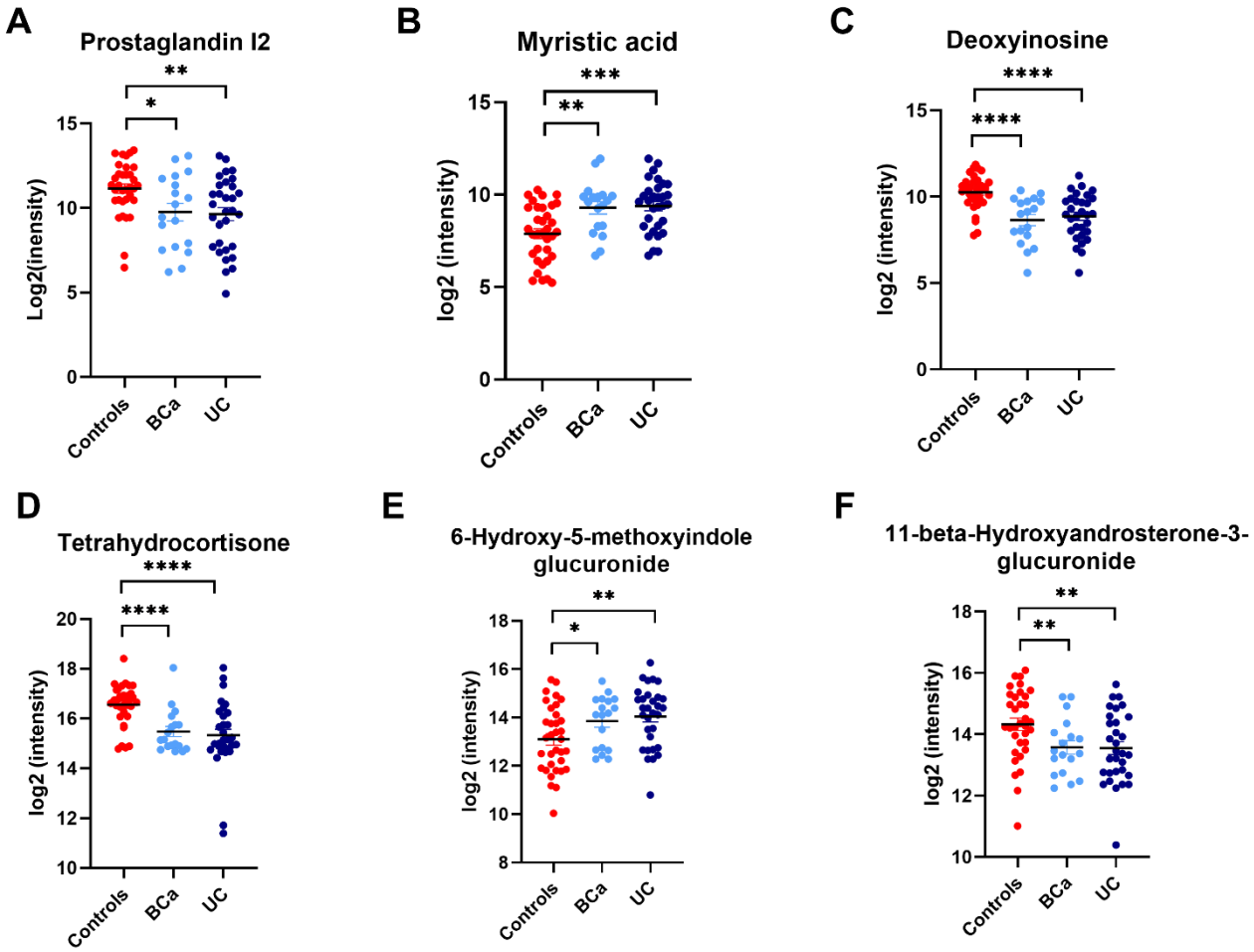
Supplementary Figure 3. | Analysis of metabolic profiling variation of UTUC and BCa without hematuria. **(A)**. Score plot of supervised PCA overview of urine metabolic profiling between UTUC and BCa without hematuria. **(B)**. Two hundred permutation tests of the OPLS-DA model based on UTUC and BCa urine metabolome. **(C).** The relative intensity of metabolites showed by heatmap. **(D).** Pathway analysis of the differential metabolites between the two subgroups.



Supplementary Figure 4. | Analysis of metabolic profiling variation of UTUC and BCa with hematuria. **(A)**. Score plot of supervised PCA overview of urine metabolic profiling between UTUC and BCa with hematuria. **(B)**. Two hundred permutation tests of the OPLS-DA model based on UTUC and BCa urine metabolome. **(C).** The relative intensity of metabolites showed by heatmap.



Supplementary Figure 5. | ROC curves for each metabolite selected for constructing the model. **(A)**.The ROC curves of metabolites in the panel when comparing UCs and HCs. **(B)**. The ROC curves of metabolites in the panel when comparing UTUC and BCa without hematuria. **(C).** The ROC curves of metabolites in the panel when comparing UTUC and BCa with hematuria.



Supplementary Figure 6. | The differential metabolites both in BCa vs controls and in UC vs controls. **(A).** The intensity of prostaglandin I2 **(B).** The intensity of mystric acid **(C).** The intensity of deoxyinosine **(D).** The intensity of Tetrahydrocortisone **(E).** The intensity of 6-hydroxy-5-methoxyindole glucuronide **(F).** The intensity of 11-beta-hydroxyandrosterone-3-glucuronide